

>> Core Language

Language	Version	Role
Python 3	3.x	Primary server-side language for the entire backend
Dart	3.x	Worker App & Resident App (compiled / Flutter)
JavaScript	ES6+	Admin web panel frontend logic
C++ (Arduino)	N/A	ESP32 microcontroller firmware for IoT sensors
HTML5 / CSS3	N/A	Admin web panel structure & styling

* Backend Framework & Libraries

Library	Purpose
Flask	Core web framework -- HTTP routing, static files, API Gateway
Flask-JWT-Extended	JWT Bearer Token authentication for all protected endpoints
Flask-Limiter	Rate limiting -- 5,000 req/hour per IP (brute-force protection)
Flask-CORS	Cross-Origin Resource Sharing -- allows Flutter/web apps to connect
Werkzeug	Secure password hashing (bcrypt via generate/check_password_hash)

Database

Technology	Type	Details
SQLite 3	Relational DB	File-based SQL database (waterhall.db)
Python sqlite3	Built-in module	Raw SQL queries -- no ORM, full control

Database Tables:

users	Admin & Worker accounts with hashed passwords
households	Registered resident household records
puroks	Barangay zone/purok assignments
water_meters	Meter serial numbers and last readings per household
billing_records	Monthly water consumption billing history
reservoir_quality_readings	IoT sensor data: water level, turbidity, pH
maintenance_logs	Field worker maintenance activity logs
announcements	Barangay announcements broadcast to residents
flow_readings	Flow rate data per purok from IoT sensors

! Security & DevSecOps

Feature	Implementation
Authentication	JWT Bearer Tokens (HS256) -- 24-hour expiry
Password Storage	Bcrypt hashing via Werkzeug -- no plaintext passwords stored
Rate Limiting	Flask-Limiter: 5,000 req/hr, login endpoint limited to 5/min
CORS Policy	Flask-CORS allowing trusted app origins only
Cache-Control	CDN-friendly: static assets cached 3600s; APIs = no-store
API Gateway	Single Flask server routes requests to all portals

@ Frontend & Client-Side Technologies

Layer	Technology	Details
Admin Web Panel	HTML5 + CSS3 + JavaScript	Glassmorphism UI, served by Flask
Admin Web Panel	Chart.js (CDN)	Live data visualization charts
Worker App	Dart -> JavaScript (dart2js)	Compiled Dart running in browser
Resident Mobile	Flutter (Dart)	Native Android APK
Typography	Google Fonts -- Outfit	Modern premium sans-serif

~ IoT Hardware & Communication Layer

Component	Technology	Protocol
Microcontroller	ESP32	WiFi + HTTP POST
Water Level	Ultrasonic / Float Sensor	HTTP to /api/central-assets/update
Turbidity	Turbidity Sensor (NTU)	HTTP to /api/central-assets/update
pH Level	pH Probe Sensor	HTTP to /api/central-assets/update
Firmware Language	C++ (Arduino Framework)	Embedded

= Full Tech Stack Summary

Server Language	Python 3
Web Framework	Flask
Authentication	Flask-JWT-Extended (JWT/OAuth-ready)
Rate Limiting	Flask-Limiter



WATERHALL System

Backend Technologies & Programming Languages

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Cross-Origin	Flask-CORS
Password Security	Werkzeug (bcrypt)
Database	SQLite 3 (via Python sqlite3 module)
Admin Panel	HTML5 + Vanilla CSS + Vanilla JS
Data Charts	Chart.js
Worker App	Dart (compiled to JS via dart2js)
Resident Mobile App	Flutter + Dart (Android APK)
IoT Firmware	C++ on ESP32 (Arduino framework)